

Problem 1. Find the volume under the surface $z = \frac{\ln x \ln y}{4xy}$ and above the rectangular region $1 \leq x \leq e^2$, $1 \leq y \leq e^2$.

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$$\begin{aligned} & \frac{1}{4} \int_1^{e^2} \int_1^{e^2} \frac{\ln x}{x} \frac{\ln y}{y} dx dy \\ &= \frac{1}{4} \left(\int_1^{e^2} \frac{\ln x}{x} dx \right) \left(\int_1^{e^2} \frac{\ln y}{y} dy \right) \end{aligned}$$

$$\text{Let } \ln x = u$$

$$\frac{dx}{x} = du$$

$$x = 1 \Rightarrow u = 0$$

$$x = e^2 \Rightarrow u = \ln e^2 = 2$$

$$\text{Let } \ln y = v$$

$$\frac{dy}{y} = dv$$

$$y = 1 \Rightarrow v = 0$$

$$y = e^2 \Rightarrow v = \ln e^2 = 2$$

$$\frac{1}{4} \int_1^{e^2} \int_1^{e^2} \frac{\ln x}{x} \frac{\ln y}{y} dx dy$$

$$= \frac{1}{4} \int_0^2 u du \int_0^2 v dv$$

$$\begin{aligned} &= \frac{1}{4} \left(\frac{u^2}{2} \right) \Big|_0^2 \left(\frac{v^2}{2} \right) \Big|_0^2 = \frac{1}{4} \left(\frac{4}{2} - 0 \right) \left(\frac{4}{2} - 0 \right) \\ &= \frac{1}{4} (2)(2) = 1 \end{aligned}$$

Problem 2. If X is a normal random variable with parameters $\mu = 1$ and $\sigma^2 = 9$. Find

(a) $P(4 < X < 7)$

$$\mu = 1 \quad \sigma = 3$$

(b) $P(|X - 1| > 6)$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

(a) $4 < X < 7$

$$\frac{4-1}{3} < \frac{X-\mu}{\sigma} < \frac{7-1}{3}$$

$$1 < Z < 2$$

(1)

By 68-95-99.7 rule we have

(1) $P(-1 < Z < 1) = 0.68$

by symmetry

(0.5)

$$\Rightarrow P(0 < Z < 1) = \frac{0.68}{2}$$

(1) $P(-2 < Z < 2) = 0.95$

by symmetry

(0.5)

$$\Rightarrow P(0 < Z < 2) = \frac{0.95}{2}$$

$$P(1 < Z < 2) = P(0 < Z < 2) - P(0 < Z < 1) \quad (1)$$

$$= \frac{0.95 - 0.68}{2} = \frac{0.27}{2} \quad (0.5)$$

(b) $P(|X - 1| > 6) = P(|X - \mu| > 2\sigma) = P\left(\frac{|X - \mu|}{\sigma} > 2\right)$

$$= 1 - P\left(\frac{|X - \mu|}{\sigma} \leq 2\right) \quad (1)$$

$$= 1 - P(|Z| \leq 2)$$

$$= 1 - P(-2 < Z < 2) \quad (1)$$

$$(1) = 1 - 0.95 = 0.05 \quad (0.5)$$

For continuous
random variables
one point doesn't
matter

by 68-95-99.7 rule

Problem 3. Recall we say $X \sim \text{Exp}(\lambda)$ when the pdf $f(x)$ is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Assume the length of a phone call is exponentially distributed with parameter $\lambda = \frac{1}{4}$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you have to wait

(a) more than 4 minutes;

(b) between 4 and 8 minutes;

let $T = \text{wait time at public booth} \sim \text{EXP}(\frac{1}{4})$

$$\begin{aligned} \text{(a)} \quad P(T > 4) &= \int_4^{\infty} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_4^{\infty} \\ &= e^{-4\lambda} = e^{-\frac{4}{4}} = e^{-1} = \frac{1}{e} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(4 < T < 8) &= \int_4^8 \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_4^8 \\ &= -e^{-8\lambda} + e^{-4\lambda} \\ &= -e^{-2} + e^{-1} \\ &= \frac{1}{e} - \frac{1}{e^2} \end{aligned}$$